Esp32Tap Full-Breadboard Netlist

Hardware: ESP32-S3 DevKitC-1 N8R8 · Omron G5V-2 DC5 · TPS3700 · TPS70950 · SN74AHC08N · SN74AHC126N

Map both RJ45 breakouts with continuity before wiring. Leave COIL POWER removed. The currently flashed diagnostic firmware cannot energize the relay or transmit to the treadmill.

1. RJ45 and relay contacts

RJ45 pin	Console side	Motor side	Function
1	CONSOLE.1 → GND	MOTOR.1 → GND	Ground pass-through
2	CONSOLE.2 → +8V_RAW	MOTOR.2 → +8V_RAW	+8 V pass-through
3	CONSOLE.3 ↔ MOTOR.3; 10 kΩ tap to GPIO16		Direct pass-through / receive tap
4	CONSOLE.4 ↔ MOTOR.4		Direct pass-through
5	CONSOLE.5 ↔ MOTOR.5		Direct safety pass-through
6	CONSOLE.6 → K1 pin 4	MOTOR.6 → K1 pin 6	Relay NC bypass; K1 pin 8 is ESP TX
7	CONSOLE.7 → GND	MOTOR.7 → GND	Second ground
8	CONSOLE.8 → +8V_RAW	MOTOR.8 → +8V_RAW	Second +8 V

G5V-2 pin	Connection	Function
1	RELAY_COIL+	5 V coil
16	RELAY_COIL− / BC337 collector	Coil low side
4	CONSOLE.6	NC
6	MOTOR.6	COM
8	TX_DRV	NO
13	GPIO4 + 10 kΩ pull-up	NC feedback
11	GND	Feedback COM
9	GPIO5 + 10 kΩ pull-up	NO feedback

The relay datasheet numbers are a **bottom view**. Meter-verify the coil and both contact poles. Coil off must give K1 4↔6 continuity and K1 8↛6 open.

2. Protected power

From	Through	To	Notes
+8V_RAW	RXEF075	+8V_F	Fuse only the local electronics tap
+8V_F	1N5822 anode → cathode/stripe	VIN	Reverse-polarity protection
VIN	P6KE12A cathode/stripe	P6KE12A anode → GND	Transient clamp

TSR1 pin	Net
1	VIN

Capacitor	Connection
22 µF	VIN to GND

TSR1 pin	Net
2	GND
3	TSR_3V3

STANDALONE POWER jumper: TSR_3V3 → LOGIC_3V3. Remove whenever UART USB is connected.

Capacitor	Connection
10 µF	VIN to GND near TSR
10 µF	TSR_3V3 to GND

3. TPS3700 voltage permission

Pin	Connection	Associated parts
1 OUTA	TREAD_OK	Tied to pin 6
2	GND	—
3 INA+	UV_SENSE	VIN—150 kΩ—node—10 kΩ—GND; 1 nF node-to-GND
4 INB−	OV_SENSE	VIN—255 kΩ—node—10 kΩ—GND; 1 nF node-to-GND
5 VDD	VIN	100 nF to GND
6 OUTB	TREAD_OK	Tied to pin 1

TREAD_OK: 10 kΩ to LOGIC_3V3; 100 kΩ to GND; through 4.7 kΩ to GPIO6.

4. SN74AHC08N

Pins	Connection
1 / 2 / 3	GPIO21 RELAY_CMD / TREAD_OK / RELAY_GATE
4 / 5 / 6	GPIO15 TX_ENABLE / TREAD_OK / TX_GATE
7 / 14	GND / LOGIC_3V3; 100 nF directly between them
9,10,12,13	GND — unused inputs must not float
8,11	Leave open — unused outputs

GPIO21 and GPIO15 each have a separate 10 kΩ pull-down to GND.

5. Relay supply and driver

TPS709 pin	Connection
1 IN	VIN; 1 µF to GND
2	GND
3 EN	RELAY_GATE
4	Leave open
5 OUT	+5V_RLY; 4.7 µF to GND

Driver connection
RELAY_GATE → 560 Ω → BC337 base
BC337 base → 10 kΩ → GND
BC337 emitter → GND
BC337 collector → G5V-2 pin 16
+5V_RLY → removable COIL POWER → G5V-2 pin 1
P6KE6.8CA directly across relay pins 1 and 16

Identify BC337 base/collector/emitter from the physical lot; do not assume a generic TO-92 order. Keep COIL POWER removed initially.

6. SN74AHC126N and UART taps

Pins	Connection
1 / 2 / 3	TX_GATE / GPIO17 / output through 100 Ω to K1 pin 8
7 / 14	GND / LOGIC_3V3; 100 nF directly between them
4,10,13 and 5,9,12	GND — unused enables and inputs
6,8,11	Leave open — unused outputs

Tap	Connection
Console serial	CONSOLE.6 → 10 kΩ → GPIO18
Pin 3	RJ45 pin 3 → 10 kΩ → GPIO16

7. Feedback, USB presence, and indicators

Function	Connection
Bypass feedback	K1 pin 13 → GPIO4; GPIO4 → 10 kΩ → LOGIC_3V3
Transfer feedback	K1 pin 9 → GPIO5; GPIO5 → 10 kΩ → LOGIC_3V3; K1 pin 11 → GND
USB presence	DevKit 5V/VBUS → 2N7000 gate and 10 kΩ to GND; source → GND; drain → GPIO7 and 10 kΩ to LOGIC_3V3
Status LED	GPIO38 → 330 Ω → green LED anode; cathode → GND
Power LED	LOGIC_3V3 → 1 kΩ → red LED anode; cathode → GND

Feedback truth: relay off/bypass = GPIO4 0, GPIO5 1. Relay energized = GPIO4 1, GPIO5 0. Treat 00 and 11 as faults.

8. DevKit header summary

DevKit pin	Net / purpose
GND	GND
3V3	LOGIC_3V3
GPIO4	K1 NC feedback
GPIO5	K1 NO feedback
GPIO6	TREAD_OK through 4.7 kΩ
GPIO7	VBUS_PRESENT_N
GPIO15	TX_ENABLE
GPIO16	RJ45 pin-3 receive
GPIO17	Buffered motor TX
GPIO18	Console pin-6 receive
GPIO21	RELAY_CMD
GPIO38	Green status LED